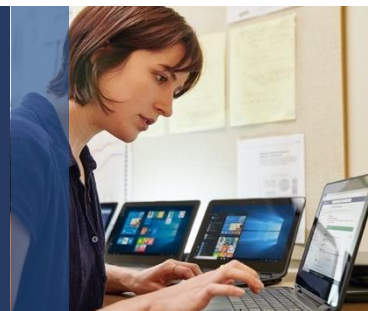




AI-900T00A Microsoft Azure AI Fundamentals

COURSE DATASHEET



Course description

Artificial Intelligence (AI) will define the next generation of software solutions and unlocks the potential to create amazing applications that improve life for everyone. This course introduces AI and the Microsoft services students can use to create AI solutions.

Audience

The Azure AI Fundamentals course is designed for anyone interested in learning about the types of solutions AI makes possible and the services on Microsoft Azure that are used to create them.

Students don't need to have any experience with Microsoft Azure before taking this course; however, a basic level of familiarity with computer technology, cloud computing, and the Internet is assumed. Some of the concepts covered in the course require a basic understanding of mathematics, such as the ability to interpret charts. The course includes hands-on activities that involve working with data and running Python, so knowledge of fundamental programming principles will be helpful.

The course materials available from Microsoft Learn for Educators are optimized for higher education instructors or instructors that teach in similar education settings (for example, early college programs in secondary schools, workforce training programs, and adult education, or continuing education programs). The Microsoft Official Course and certification exam will be localized into French, German, Japanese, Korean, Simplified Chinese, and Spanish.

Learning objectives

After completing this course, students will be able to:

- Describe Artificial Intelligence workloads and considerations
- Describe fundamental principles of machine learning on Azure
- Describe features of computer vision workloads on Azure
- Describe features of Natural Language Processing (NLP) workloads on Azure
- Describe features of conversational AI workloads on Azure

See the full [certification skills outline](#) for further details.

Exam Relationship

This course is directly mapped to and supports learning for the AI-900 Microsoft Azure AI Fundamentals certification exam. The table below outlines the relative weight percentages for each area of the exam. The higher the percentage, the more questions students are likely to see in that area.

Prior to April 29, 2022:

STUDY AREAS	WEIGHTS
Describe Artificial Intelligence workloads and considerations	15-20%
Describe fundamental principles of machine learning on Azure	30-35%
Describe features of computer vision workloads on Azure	15-20%
Describe features of Natural Language Processing workloads on Azure	15-20%
Describe features of conversational AI workloads on Azure	15-20%

After April 29, 2022:

STUDY AREAS	WEIGHTS
Describe Artificial Intelligence workloads and considerations	20-25%
Describe fundamental principles of machine learning on Azure	25-30%
Describe features of computer vision workloads on Azure	15-20%
Describe features of natural language processing workloads on Azure	25-30%

See the full [certification skills outline](#) for further details.

Course Materials

In addition to this course datasheet the following course materials are available through [Microsoft Learn for Educators](#):

- **Microsoft Official Course:** Four, stand-alone PowerPoints that can be delivered in 60-105 minutes of class time. The presentations are sub-divided into shorter sections and facilitated through PowerPoint slide decks that include lecture topics, instructor notes, demonstrations, and hands-on activities.
- **Educator Teaching Guide:** Guide for educators who plan to implement the course in academic settings. It includes detailed explanations of the course materials, technical requirements, and setup. It also includes recommended resources for preparing for the certification exam, accommodating students, and remote teaching.
- **Online Training:** Self-paced online learning content on Microsoft Learn that aligns directly to Exam AI900: Microsoft Azure AI Fundamentals and to the Microsoft Official Course. Each presentation stands alone and includes explanations of technical concepts, hands-on activities, and knowledge checks. This course includes the following six Learning Paths:
 - [Microsoft Azure AI Fundamentals: Get started with artificial intelligence](#)
 - [Microsoft Azure AI Fundamentals: Explore visual tools for machine learning](#)
 - [Microsoft Azure AI Fundamentals: Explore computer vision](#)
 - [Microsoft Azure AI Fundamentals: Explore natural language processing](#)
 - [Microsoft Azure AI Fundamentals: Explore decision support](#)
 - [Microsoft Azure AI Fundamentals: Explore knowledge mining](#) (*material not in scope for exam*)

- **Reference Delivery Videos:** On-demand recordings of an experienced instructor delivering each course presentation. Note that these are only available to instructors preparing to teach the course and cannot be shared with students.
- **On-demand Train the Trainer Session (coming soon!):** Recorded presentation that provides tips and best practices for delivering the course.

Flexible Format

The course is designed to enable higher education instructors to deliver a complete subject from the Microsoft Official Course during 1-2 class meetings and then direct students to complete the aligned online learning on Microsoft Learn. However, the content and online learning have a flexible design that includes stand-alone units. This allows instructors to easily modify the delivery format, supplement the content, or integrate the course with other curricula.

Recommended Instructor Preparation

Instructors are strongly encouraged to review the Azure AI Fundamentals Learning Paths on Microsoft Learn and complete Exam AI-900: Microsoft Azure AI Fundamentals prior to teaching the course to verify their understanding of the course content and gain insight into how to support students preparing for the exam. Instructors are also encouraged to prepare to support students who are developing their understanding of the basic concepts and skills listed in the Audience section of this document.



Appendix A

The following table provides a snapshot of the presentations included in the instructor-led course content, detailed timing, and a link to the aligned segments of the Online Training on Microsoft Learn:

Microsoft Official Course		Aligned Online Training on Microsoft Learn
Course Introduction		
10 minutes		
Introduction to AI 80 minutes	- Introduction to Artificial Intelligence - Artificial Intelligence in Microsoft Azure Lab: Explore Cognitive Services	Microsoft Azure AI Fundamentals: Get started with artificial intelligence 35 minutes
Machine Learning 105 minutes	- Introduction to Machine Learning - Azure Machine Learning Lab: Explore Machine Learning	Microsoft Azure AI Fundamentals: Explore visual tools for machine learning Technologies used in exercises: - Azure Machine Learning Automated Machine Learning - Azure Machine Learning designer 3 hours 30 minutes
Computer Vision 90 minutes	- Computer Vision Concepts - Computer Vision solutions in Azure Lab: Explore Computer Vision	Microsoft Azure AI Fundamentals: Explore computer vision Technologies used in exercise: - Computer Vision Service - Custom Vision Service - Face Service - Form Recognizer Service 3 hours
Natural Language Processing 90 minutes	- Introduction to Natural Language Processing - Building Natural Language Solutions in Azure Lab: Explore text analytics	Microsoft Azure AI Fundamentals: Explore natural language processing Technologies used in exercises: - Text Analytics feature of the Language Service - Conversational Language Understanding feature of the Language Service - Speech Service - Translator Service - Custom Question Answering feature of the Language Service - Azure Bot Service 2 hours 42 minutes

*Note that the Timings are for delivery in a higher education setting without supplemental instruction about related concepts. Instructors who want to deliver in other settings or provide supplemental instruction in foundational or related concepts should plan for longer implementation.

Appendix B

Extended Outline

This outline provides detailed information about the presentations included in the instructor-led course content and the aligned segments of the Online Training on Microsoft Learn.

Subject and description	Section	Key topics	In-class demonstrations and labs	Aligned Online Training on Microsoft Learn
Course Introduction This presentation provides a high-level course format and objectives as well as an outline of the course.				
Introduction to AI In this presentation, you'll learn about common uses of artificial intelligence (AI), and the different types of workloads associated with AI. You'll then explore considerations and principles for responsible AI development.	Introduction to Artificial Intelligence	1. What is Artificial Intelligence? 2. Common AI Workloads 3. Principles of Responsible AI	Demo: Getting started with the Azure portal	Microsoft Azure AI Fundamentals: Get started with artificial intelligence
	Artificial Intelligence in Microsoft Azure	1. Azure Basics 2. AI in Microsoft Azure 3. Cognitive Services	Lab: Explore Cognitive Services	
Machine Learning Machine learning is the foundation for modern AI solutions. In this presentation, you'll learn about some fundamental machine learning concepts, and how to use the Azure Machine Learning service to create and publish machine learning models.	Introduction to Machine Learning	1. What is machine learning? 2. Types of machine learning 3. Model training and validation		Microsoft Azure AI Fundamentals: Explore visual tools for machine learning
	Azure Machine Learning	1. What is Azure Machine Learning? 2. Automated machine learning 3. Azure Machine Learning Designer	Lab: Explore Machine Learning Optional Demo: Create a classification model with Azure Machine Learning Designer	
Subject and description	Section	Key topics	In-class demonstrations and labs	Aligned Online Training on Microsoft Learn

Computer Vision Computer vision is the area of AI that deals with understanding the world visually, through images, video files, and cameras. In this presentation you'll explore multiple computer vision techniques and services	Computer Vision Concepts	1. What is Computer Vision 2. Applications of Computer Vision 3. Azure Computer Vision	Demo: Show some examples of Computer Vision	Microsoft Azure AI Fundamentals: Explore computer vision
	Creating Computer Vision solutions in Azure	1. Image analysis with the Computer Vision Service 2. Custom Vision 3. Face Service 4. Computer Vision Text 5. Form Recognizer	Lab: Explore Computer Vision	
Natural Language Processing This presentation describes scenarios for AI solutions that can process written and spoken language. You'll learn about Azure services that can be used to build solutions that analyze text, recognize, and synthesize speech, translate between languages, and interpret commands	Introduction to Natural Language Processing	1. What is Natural Language Processing? 2. What is Conversational AI? 3. Natural Language Processing in Azure	Demo: Show some examples of NLP	Microsoft Azure AI Fundamentals: Explore natural language processing
	Building Natural Language Solutions in Azure	1. Analyzing Text 2. Speech Recognition and Synthesis 3. Translation 4. Conversational Language Understanding 5. Question Answering 6. Azure Bot Service	Lab: Explore text analytics	